Bike Price Prediction Using Machine Learning: A Comparative Evaluation of Regression Models

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***Abstract*—Accurate prediction of used-bike selling prices is an important machine learning problem because vehicle prices depend on multiple interacting factors such as vehicle age, accumulated distance, original ex-showroom price, ownership history, seller type, and vehicle model. Manual estimation of used-bike prices can be subjective and may not consistently reflect the relationships present in historical market data.**

This research presents a machine learning based regression system for predicting the selling price of used bikes. The study uses a bike-price dataset containing information about bike name, selling price, manufacturing year, seller type, ownership status, kilometers driven, and ex-showroom price. The original dataset contains 1,061 records and seven variables. Six duplicate records were identified and removed, resulting in 1,055 usable records. Missing ex-showroom-price values were handled using group-wise median imputation followed by global median imputation where necessary. Additional features including bike age, kilometers driven per year, log-transformed kilometers driven, and log-transformed ex-showroom price were constructed.

Five regression algorithms were experimentally evaluated: Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and XGBoost. A preprocessing pipeline was used to impute missing values and encode categorical variables using one-hot encoding. The dataset was divided into 80% training data and 20% testing data using a fixed random state. Among the evaluated models, XGBoost achieved the best performance with an R^2 score of 0.9143, mean absolute error (MAE) of 11,763.92, and root mean squared error (RMSE) of 19,353.69. The results demonstrate that gradient-boosted decision trees can effectively capture nonlinear relationships in used-bike pricing data and provide a strong baseline for automated price estimation.

***Index Terms*—Bike Price Prediction, Machine Learning, Regression, XGBoost, Random Forest, Gradient Boosting, Feature Engineering, Used Vehicle Valuation, Predictive Analytics**

I. Introduction

The used two-wheeler market contains a large number of vehicles with different models, manufacturing years, ownership histories, usage patterns, and original prices. Consequently, estimating a reasonable resale price for a used bike is a non-linear problem influenced by several numerical and categorical factors.

Traditional pricing methods commonly depend on manual inspection, seller expectations, vehicle condition, historical experience, and market knowledge. Although these approaches can be useful, they may introduce inconsistencies because different sellers or buyers can assign different values to the same vehicle.

Machine learning provides a data-driven alternative to manual valuation. Regression algorithms can learn relationships between historical vehicle attributes and observed selling prices. Once a model has been trained, it can estimate the expected selling price of a previously unseen bike based on its available characteristics.

This work investigates several machine learning regression algorithms for used-bike price prediction. The study compares Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and XGBoost. The objective is not only to obtain a prediction but also to determine which model provides the best balance of prediction accuracy and generalization on unseen test data.

The implementation uses Python together with Pandas, NumPy, Matplotlib, and Scikit-learn. XGBoost is additionally used as a gradient-boosting implementation for the final predictive model.

II. Problem Statement

Used-bike selling prices are affected by multiple variables whose relationships are not necessarily linear. A bike with a newer manufacturing year may generally have a higher price, while high kilometer usage may decrease its value. However, the magnitude of these effects can vary considerably between different bike models.

The problem addressed in this research is therefore formulated as:

> Given the historical attributes of a used bike, develop a machine learning regression model capable of predicting its expected selling price accurately on previously unseen observations.

The target variable is the *selling price*. The available predictor variables include bike name, year, seller type, ownership, kilometers driven, and ex-showroom price, together with engineered features.

III. Research Objectives

The major objectives of this research are:

1) To analyze a historical used-bike price dataset.

2) To identify numerical and categorical variables relevant to selling-price prediction.
3) To detect and handle duplicate records and missing values.
4) To construct meaningful derived features from existing vehicle attributes.
5) To encode categorical variables into a machine-learning compatible representation.
6) To develop multiple regression models for price prediction.
7) To compare the models using R^2, MAE, and RMSE.
8) To identify the best-performing regression algorithm.
9) To evaluate the selected model on unseen test observations.
10) To establish a reproducible machine-learning workflow for used-bike price estimation.

IV. Research Questions

This research investigates the following questions:

1) Can machine learning accurately predict used-bike selling prices from historical vehicle information?
2) Which regression algorithm performs best for the given dataset?
3) How does feature engineering affect the representation of vehicle age and usage?
4) How effectively can categorical variables such as bike model, seller type, and ownership be incorporated into the predictive pipeline?
5) Does a nonlinear boosting model provide better predictive performance than conventional linear and ensemble regression methods?

V. Literature Review / Related Work

Machine learning based regression has been widely applied to automobile and vehicle-price prediction problems. Regression algorithms attempt to learn a function that maps a set of input variables to a continuous target value.

Linear Regression is one of the simplest regression methods and assumes an approximately linear relationship between the predictors and target variable. It provides a useful baseline because its implementation and interpretation are relatively straightforward [5].

Tree-based ensemble methods provide an alternative when relationships between variables are nonlinear. Random Forest constructs multiple decision trees using randomized subsets of observations and features and aggregates their predictions [1].

Gradient Boosting builds an additive predictive model by sequentially training weak learners to reduce prediction error. The approach is particularly useful for structured tabular data [2].

Extra Trees introduces additional randomization into tree construction and can capture nonlinear relationships without requiring explicit mathematical transformations of every interaction.

XGBoost is an optimized gradient-boosting framework designed for efficient and scalable predictive modeling. Its regularization, tree-based learning, and gradient-boosting strategy make it effective for many structured-data regression tasks [3].

Scikit-learn provides a consistent framework for preprocessing, pipeline construction, model fitting, and evaluation [4].

VI. Research Gap

Although vehicle-price prediction is a common machine-learning application, the effectiveness of different regression algorithms can vary depending on the characteristics of a particular dataset.

A practical gap exists between simply applying a single regression algorithm and systematically comparing multiple models under a consistent preprocessing framework.

This study addresses this gap by:

- using both numerical and categorical vehicle attributes;
- applying systematic missing-value handling;
- creating domain-relevant engineered features;
- applying one-hot encoding to categorical attributes;
- evaluating multiple regression algorithms under the same train-test split;
- comparing models using multiple regression metrics; and
- selecting the final model based on observed predictive performance.

The study therefore provides an experimental comparison of classical and ensemble regression approaches for the specific used-bike pricing dataset.

VII. Dataset Description

The dataset used in this research is the `BIKE DETAILS.csv` dataset. It contains historical information about used bikes and their selling prices.

The original dataset contains 1,061 records and seven columns.

TABLE I
Dataset Attributes

Attribute	Description
name	Name/model of the bike
selling_price	Target selling price
year	Manufacturing year
seller_type	Type of seller
owner	Ownership category
km_driven	Distance driven by the bike
ex_showroom_price	Original ex-showroom price

The target variable is:

$$y = \text{selling_price}$$

The original dataset contains 435 missing values in the `ex_showroom_price` column. The other primary columns do not contain missing values in the initial inspection [9].

The most frequent bike model in the dataset is Bajaj Pulsar 150, which occurs 41 times. The seller type is dominated by individual sellers, while the most frequent ownership category is first-owner vehicles.

VIII. Methodology

The proposed methodology consists of the following stages:

1) Dataset loading.
2) Data inspection.
3) Duplicate detection and removal.
4) Data-type conversion.
5) Missing-value treatment.
6) Feature engineering.
7) Exploratory data analysis.
8) Feature and target separation.
9) Categorical and numerical preprocessing.
10) Train-test splitting.
11) Regression model development.
12) Model evaluation.
13) Model comparison.
14) Final model selection.
15) Prediction on unseen test observations.

IX. System / ML Workflow

The complete machine-learning workflow can be represented as:

$$\text{Raw Dataset} \rightarrow \text{Data Cleaning} \rightarrow \text{Feature Engineering} \rightarrow \text{Preprocessing} \rightarrow \text{Train/Test Split} \rightarrow \text{Model Training} \rightarrow \text{Evaluation} \rightarrow \text{Final Pre} \tag{1}$$

The workflow is designed to ensure that raw categorical and numerical variables are transformed into a representation suitable for machine-learning algorithms.

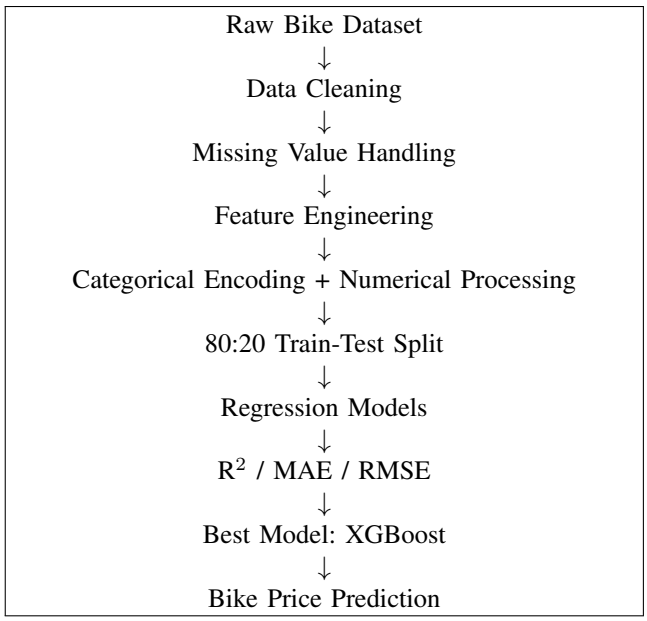


Fig. 1. Machine Learning Workflow

X. Data Preprocessing

Data preprocessing is an essential stage because machine-learning algorithms require consistent and valid input representations.

A. Duplicate Removal

The initial dataset contained six duplicate records. These records were removed using the Pandas `drop_duplicates()` operation.

$$N_{\text{after duplicates}} = 1061 - 6 = 1055$$

Thus, 1,055 records remained after duplicate removal.

B. Data-Type Conversion

The following variables were explicitly converted to numeric representations:

- year
- km_driven
- ex_showroom_price
- selling_price

Conversion was performed using numeric coercion so that invalid values could be represented as missing values for subsequent processing.

C. Target Validation

Rows with missing values in the target variable `selling_price` were removed. The resulting dataset contained 1,055 observations.

D. Categorical Missing Values

Categorical variables were assigned an "Unknown" value if a missing value was present.

E. Ex-Showroom Price Imputation

Missing ex-showroom prices were first filled using the median ex-showroom price for the corresponding bike model:

$$x_i = \text{median}\,(x_j : \text{name}_j = \text{name}_i)$$

If a model-specific median was unavailable, the overall median was used.

This process eliminated the missing values in the final modeling dataset.

XI. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure, distribution, and range of the dataset.

The selling-price distribution was visualized using a histogram. The dataset shows substantial variation in selling prices, with values ranging from 5,000 to 760,000 in the original data.

The descriptive statistics show:

TABLE II
Important Dataset Statistics

Variable	Mean	Median	Maximum
Selling Price	59638.15	45000	760000
Year	2013.87	2015	2020
KM Driven	34359.83	25000	880000
Ex-showroom Price	87958.71	72752.50	1278000

The relatively large difference between the median and maximum values of several numerical variables indicates substantial variation and the presence of high-valued observations.

A. Selling Price Distribution

A histogram was generated to visualize the frequency distribution of selling prices.

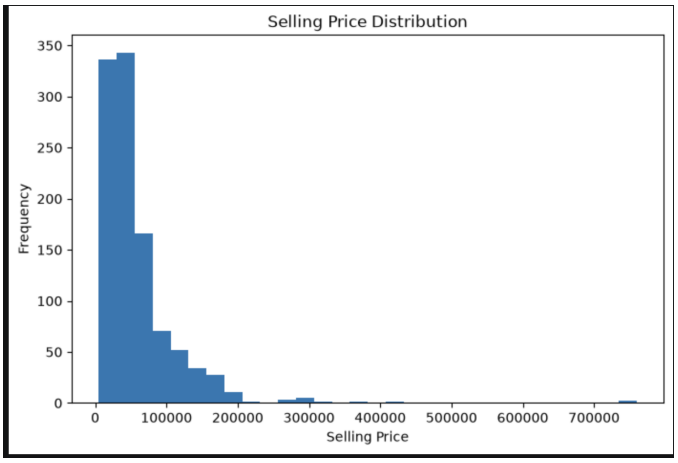


Fig. 2. Distribution of Bike Selling Prices

XII. Feature Engineering / Feature Selection

Feature engineering was performed to derive additional information from the original variables.

A. Bike Age

Bike age was calculated using:

$$\text{BikeAge} = 2026 - \text{Year} \tag{2}$$

The feature represents the approximate age of the bike relative to the year of analysis.

B. Kilometers per Year

The usage intensity was represented using:

$$\text{KMPerYear} = \frac{\text{KMDriven}}{\text{BikeAge} + 1} \tag{3}$$

The addition of one prevents division by zero for a newly manufactured vehicle.

C. Log Transformation of Kilometer Usage

Because kilometer usage can have a large range, a logarithmic transformation was generated:

$$\text{LogKM} = \log(1 + \text{KMDriven}) \tag{4}$$

D. Log Transformation of Ex-Showroom Price

Similarly:

$$\text{LogExShowroom} = \log(1 + \text{ExShowroomPrice}) \tag{5}$$

The final numerical feature set consists of:

- year
- km_driven
- ex_showroom_price
- bike_age
- km_per_year
- log_km_driven
- log_ex_showroom

The categorical features are:

- name
- seller_type
- owner

After feature engineering, the model input contains 10 predictors.

XIII. Model Development

Five regression algorithms were developed and evaluated.

A. Linear Regression

Linear Regression models the target using a weighted combination of input variables:

$$\hat{y} = \beta_0 + \sum_{i=1}^{p} \beta_i x_i \tag{6}$$

It was used as a baseline model.

B. Random Forest

Random Forest combines predictions from multiple decision trees. The final prediction can be expressed as:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t(x) \tag{7}$$

where T represents the number of trees.
The implementation used 500 trees.

C. Gradient Boosting

Gradient Boosting constructs a sequence of weak learners where subsequent learners attempt to reduce previous prediction errors.
The model used:

- 500 estimators
- learning rate = 0.03
- maximum depth = 4
- Huber loss

D. Extra Trees

Extra Trees is an ensemble of highly randomized decision trees. The implementation used 500 estimators and a maximum feature fraction of 0.8.

E. XGBoost

XGBoost was used as the primary gradient-boosting model. Its objective was configured as squared-error regression:

$$L = \sum_i (y_i - \hat{y}_i)^2 \tag{8}$$

The selected experimental configuration was:

TABLE III
XGBOOST CONFIGURATION

Parameter	Value
n_estimators	1000
learning_rate	0.03
max_depth	5
min_child_weight	2
subsample	0.85
colsample_bytree	0.85
reg_alpha	0.05
reg_lambda	1
objective	reg:squarederror
random_state	42

XIV. EXPERIMENTAL SETUP

The implementation was developed in Python using the following libraries:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- XGBoost

The notebook environment used Python 3.14.4 and XGBoost 3.4.1 during experimentation.

The final dataset contains 1,055 observations.

The dataset was divided as:

TABLE IV
TRAINING AND TESTING DATA

Set	Number of Samples
Training	844
Testing	211
Total	1055

An 80:20 split was used with `random_state=42`.

XV. MODEL VALIDATION

The primary validation strategy used in the notebook is a hold-out test-set evaluation.

The data were divided into:

$$D = D_{\text{train}} \cup D_{\text{test}}$$

where:

$$|D_{\text{train}}| = 844$$

and:

$$|D_{\text{test}}| = 211$$

The model was fitted using only the training data and subsequently evaluated using the independent test data.

It should be noted that the present implementation does not include K-fold cross-validation. Therefore, the reported metrics represent performance on the fixed 20% test partition rather than an average across multiple folds.

XVI. HYPERPARAMETER TUNING

The experiments used explicitly specified model configurations. The notebook does not implement an automated GridSearchCV or RandomizedSearchCV procedure.

For the tree-based models, the selected configurations were designed to provide sufficiently expressive ensemble models while controlling learning rates and tree complexity.

For XGBoost, the main selected parameters were:

- 1,000 boosting estimators;
- learning rate of 0.03;
- maximum tree depth of 5;
- minimum child weight of 2;
- row subsampling of 0.85;
- column subsampling of 0.85;
- L1 regularization coefficient of 0.05; and
- L2 regularization coefficient of 1.

These values constitute the final experimental configuration used for the reported XGBoost results.

For future work, automated hyperparameter optimization can be performed using GridSearchCV, RandomizedSearchCV, Bayesian optimization, or Optuna.

XVII. MODEL EVALUATION

Three regression metrics were used.

A. R^2 Score

The coefficient of determination is:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{9}$$

A value closer to 1 indicates that the model explains a larger proportion of the variation in the target variable.

B. Mean Absolute Error

MAE is defined as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{10}$$

Lower MAE indicates smaller average absolute prediction errors.

C. Root Mean Squared Error

RMSE is:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{11}$$

RMSE gives greater influence to large prediction errors.

XVIII. Model Comparison

All five models were trained using the same preprocessing pipeline and evaluated on the same test partition.

TABLE V
PERFORMANCE COMPARISON OF REGRESSION MODELS

Model	R^2	MAE	RMSE
XGBoost	0.9143	11763.92	19353.69
Gradient Boosting	0.8241	13038.14	27732.38
Linear Regression	0.7681	18007.29	31840.17
Extra Trees	0.7643	13749.03	32099.19
Random Forest	0.7261	13912.79	34601.98

XGBoost achieved the highest R^2 score and the lowest RMSE and MAE among the evaluated models.

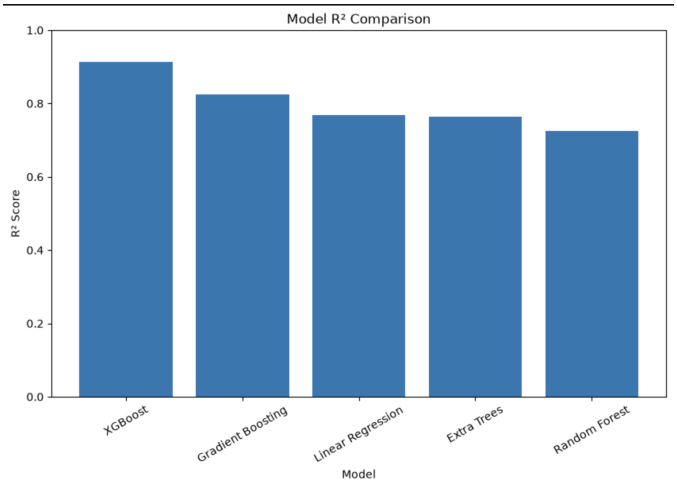


Fig. 3. Comparison of Regression Models

XIX. Final Model Selection

XGBoost was selected as the final model because it produced the best overall test-set performance.

The final model achieved:

$$R^2 = 0.9143 \tag{12}$$

$$MAE = 11763.92 \tag{13}$$

$$RMSE = 19353.69 \tag{14}$$

The R^2 result indicates that approximately 91.43% of the variance in the test-set selling prices is explained by the model.

This does not mean that every individual prediction has an accuracy of 91.43%. Instead, R^2 measures the proportion of target variance explained relative to a baseline based on the test-set mean.

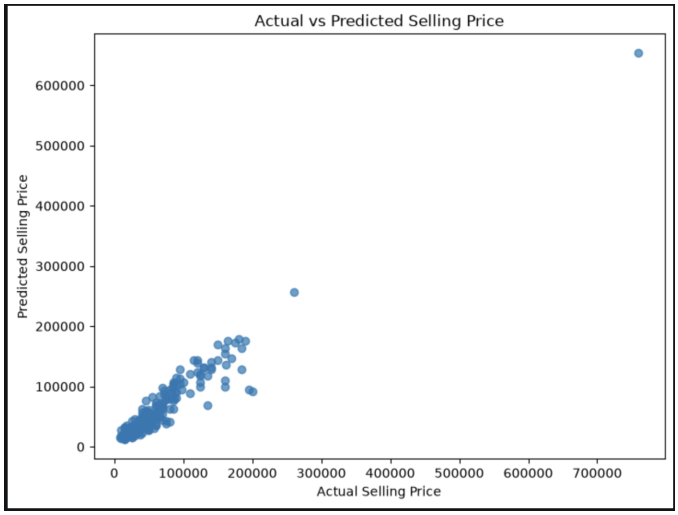


Fig. 4. Actual vs predicted selling price

XX. Sample / Unseen Data Prediction

The selected XGBoost pipeline was used to generate predictions for the 211 observations in the independent test set.

The notebook also generated a comparison table containing actual price, predicted price, and their difference.

Examples include:

TABLE VI
SAMPLE TEST-SET PREDICTIONS

Actual	Predicted	Difference
260000	257587.80	2412.20
60000	56251.65	3748.35
50000	52633.25	-2633.25
50000	53693.61	-3693.61
30000	46180.34	-16180.34
45000	36234.63	8765.38
75000	42622.45	32377.55
72000	87704.36	-15704.36
70000	55057.30	14942.70
28700	27320.32	1379.68

The prediction differences demonstrate that the model can produce close estimates for many observations while larger deviations occur for some vehicles.

XXI. Results

The final experimental results are summarized below.

TABLE VII
FINAL XGBOOST RESULTS

Metric	Value
Model	XGBoost
R^2 Score	0.9143
MAE	11,763.92
RMSE	19,353.69
Training Samples	844
Testing Samples	211

The XGBoost model clearly outperformed the other evaluated models.

The improvement in R^2 over the second-best Gradient Boosting model was:

$$0.9143 - 0.8241 = 0.0902 \tag{15}$$

Thus, XGBoost produced an approximately 0.0902 higher R^2 score on the test partition.

The corresponding relative reduction in MAE compared with Gradient Boosting was approximately:

$$\frac{13038.14 - 11763.92}{13038.14} \times 100 \approx 9.77\% \tag{16}$$

XXII. Discussion

The experimental results demonstrate that nonlinear ensemble methods can be more effective than a simple linear regression model for this dataset.

Linear Regression achieved an R^2 score of 0.7681. Although this represents meaningful predictive capability, the model assumes a linear relationship between the predictors and target. Used-bike pricing can involve interactions between model, age, mileage, ownership, and original price that are difficult to capture with a simple linear function.

Random Forest achieved an R^2 score of 0.7261. Despite being a powerful nonlinear ensemble, its performance was lower than the other boosting approaches in this experiment.

Extra Trees achieved an R^2 score of 0.7643, which was comparable to Linear Regression but considerably lower than XGBoost.

Gradient Boosting achieved a substantially stronger R^2 score of 0.8241 and lower RMSE than Linear Regression, Random Forest, and Extra Trees.

XGBoost produced the best results, achieving an R^2 score of 0.9143. The result suggests that sequential gradient boosting with the selected configuration was particularly effective at learning the complex relationships in the structured bike-price dataset.

The actual-versus-predicted comparison also indicates that the predictions generally follow the observed selling prices. However, some observations show larger errors, demonstrating that additional information about bike condition, location, service history, insurance status, engine characteristics, and market demand could potentially improve predictions.

XXIII. Limitations

Several limitations should be considered.

A. Dataset Size

The final dataset contains 1,055 observations. This is relatively small for training highly expressive machine-learning models.

B. Limited Vehicle Attributes

The dataset does not explicitly include several potentially important variables such as:

- engine displacement;
- mileage or fuel efficiency;
- engine condition;
- accident history;
- service history;
- insurance validity;
- geographic location;
- tire condition;
- number of previous repairs; and
- cosmetic condition.

These variables can influence the actual market value of a used bike.

C. Hold-Out Validation

The current experiment uses a single 80:20 train-test split. Consequently, the reported metrics may depend on the particular random partition.

K-fold cross-validation would provide a more robust estimate of generalization performance.

D. Manual Hyperparameter Configuration

The current notebook uses explicitly selected model parameters rather than an automated hyperparameter optimization procedure.

E. Temporal Considerations

Bike prices can change over time because of inflation, new-model releases, supply and demand, and market conditions. A model trained on historical records may therefore require periodic retraining.

XXIV. Conclusion

This research developed and evaluated a machine-learning system for predicting used-bike selling prices.

The study began with a dataset containing 1,061 records and seven original attributes. After duplicate removal and preprocessing, 1,055 records were used for machine-learning experiments.

Several preprocessing and feature-engineering operations were performed, including missing-value treatment, categorical encoding, bike-age calculation, kilometers-per-year calculation, and logarithmic transformations.

Five regression models were evaluated: Linear Regression, Random Forest, Gradient Boosting, Extra Trees, and XGBoost.

Among the evaluated algorithms, XGBoost achieved the best test-set performance with:

$$R^2 = 0.9143$$

$$MAE = 11763.92$$

$$RMSE = 19353.69$$

The results indicate that XGBoost can effectively model the nonlinear relationships present in the used-bike pricing data.

The research demonstrates that machine learning can provide a useful data-driven approach to automated bike price estimation. However, further improvements are possible through additional features, larger datasets, robust cross-validation, and automated hyperparameter optimization.

XXV. Future Scope

Several improvements can be explored in future work.

1) Collect a larger and more geographically diverse dataset.
2) Include additional vehicle attributes such as engine capacity, mileage, condition, insurance status, service history, and accident history.
3) Apply K-fold cross-validation to obtain more reliable generalization estimates.
4) Perform automated hyperparameter optimization using GridSearchCV, RandomizedSearchCV, Bayesian optimization, or Optuna.
5) Investigate advanced boosting algorithms such as LightGBM and CatBoost.
6) Perform feature-importance and explainability analysis using SHAP.
7) Develop a web-based price prediction application.
8) Add geographic market information to capture regional differences in vehicle prices.
9) Introduce temporal retraining so that the model adapts to changing market prices.
10) Investigate uncertainty estimation so that predictions can be accompanied by confidence intervals or prediction ranges.

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[9] Bike Price Prediction Dataset, `BIKE DETAILS.csv`, experimental dataset used in this research.

Appendix A
Implementation Details

The following pseudocode summarizes the implemented machine-learning pipeline.

Listing 1. Simplified Machine Learning Pipeline

```
Load BIKE DETAILS.csv

Remove duplicate rows

Convert numerical columns to numeric

Remove rows with missing target values

Fill missing categorical values

Calculate:
    bike_age
    km_per_year
    log_km_driven
    log_ex_showroom

Fill missing ex_showroom_price
using model-wise median

Separate:
    X = features
    y = selling_price

Define numerical features

Define categorical features

Numerical preprocessing:
    Median imputation

Categorical preprocessing:
    Most frequent imputation
    One-hot encoding

Split data:
    80% training
    20% testing

Train:
    Linear Regression
    Random Forest
    Gradient Boosting
    Extra Trees
    XGBoost

Predict on test set

Calculate:
    R2
    MAE
    RMSE

Select model with highest R2

Generate final predictions
```

Appendix B
Final XGBoost Configuration

Listing 2. Final XGBoost Configuration

```
XGBRegressor(
    n_estimators=1000,
    learning_rate=0.03,
    max_depth=5,
    min_child_weight=2,
    subsample=0.85,
    colsample_bytree=0.85,
    reg_alpha=0.05,
    reg_lambda=1,
    objective="reg:squarederror",
    random_state=42,
    n_jobs=-1
)
```

APPENDIX C
FINAL FEATURE CONFIGURATION

TABLE VIII
FINAL FEATURES USED FOR MODELING

Type	**Features**
Categorical	name
	seller_type
	owner
Numerical	year
	km_driven
	ex_showroom_price
	bike_age
	km_per_year
	log_km_driven
	log_ex_showroom

APPENDIX D
FINAL EXPERIMENTAL SUMMARY

TABLE IX
COMPLETE EXPERIMENTAL SUMMARY

Property	**Value**
Original Records	1061
Duplicates Removed	6
Final Records	1055
Original Features	6 predictors + target
Engineered Features	4
Final Predictors	10
Training Samples	844
Testing Samples	211
Models Evaluated	5
Best Model	XGBoost
Best R^2	0.9143
Final MAE	11763.92
Final RMSE	19353.69